

Decent Coaching Classes
X SB
Academic Conquest - III
ANSWER KEY (Geography — Chapters 1 to 4)

Q.1 (A) Choose the correct alternative:

- (1) (b) 6°45' N
- (2) (c) Siwaliks
- (3) (b) Venezuela
- (4) (b) North-East Trade winds

Q.1 (B) Fill in the blanks:

- (1) Thar Desert
- (2) Sao Francisco
- (3) Monsoon

Q.1 (C) True or False:

- (1) True — The Aravalis are the oldest fold mountains of the Indian Peninsula, lying in the north of the plateau region.
- (2) False — Brazil has a coastline along the Atlantic Ocean, not the Pacific Ocean.
- (3) True — Near the equator in Brazil, winds move vertically and the convergence zone of trade winds is weak, so cyclones rarely form and tropical cyclones rarely visit Brazil's coasts.

Q.2 Answer the following questions in brief (Any Four): [2 marks each]

(1) Latitudinal and longitudinal extent of India and Brazil:

India: India extends from about 8°4' N to 37°6' N latitude and from 68°7' E to 97°25' E longitude. The mainland extends between 8°4' N to 37°6' N and 68°7' E to 97°25' E, while Indira Point (southernmost tip) lies at 6°45' N.

Brazil: Brazil extends from about 5°15' N to 33°45' S latitude and from 34°47' W to 73°48' W longitude. Some part of Brazil lies in the Northern Hemisphere while most of it lies in the Southern Hemisphere; it lies entirely in the Western Hemisphere.

(2) Items to be carried during a field visit:

Personal luggage, I-Card, backpack, water bottle, notebook, pen/pencil, first-aid box, map/atlas of the region, binoculars, camera, cap, compass, and a questionnaire (if survey work is to be done).

(3) Major water divides of India:

The Aravalis act as a water divide between rivers flowing towards the Arabian Sea (like Luni) and those flowing east. The Vindhyas act as a water divide between the Ganga basin (north) and the Narmada-Tapi basin (south). The Western Ghats form the major water divide of the Peninsula, separating west-flowing rivers (meeting the Arabian Sea, e.g., Narmada, Tapi) from east-flowing rivers (meeting the Bay of Bengal, e.g., Godavari, Krishna, Kaveri).

(4) Importance of Himalayas and Indian Ocean for India's climate:

The Himalayas act as a climatic barrier — they obstruct cold winds blowing from Central Asia in winter, protecting India from extreme cold, and they cause the South-West Monsoon winds to retreat from the Shiwalik and Himachal ranges. They also cause orographic rainfall as monsoon winds strike them. The Indian Ocean, being a large water body, heats up and cools down differently from land, creating pressure differences (low pressure over Punjab/Thar desert in summer) that draw in the moisture-laden monsoon winds from the ocean, which bring rainfall to India.

(5) Formation of swamps in Pantanal:

Pantanal lies in the southwest part of the Brazilian Highlands in a low-lying, extensive continental (interior) location. Due to its flat terrain, poor natural drainage, and confluence of river systems (Paraguay and its tributaries) in this basin-like area, water accumulates instead of draining away quickly, and combined with seasonal heavy rainfall, this leads to extensive swamps and marshes.

Q.3 Give geographical reasons (Any Three): [2 marks each]

(1) There are no west-flowing rivers in Brazil:

The general slope of Brazil's physiography is towards the north, east and south-east due to the arrangement of the Guyana Highlands, Amazon Basin and Brazilian Highlands, which slope towards the Atlantic Ocean in the east. Since there is no highland/water divide sloping towards the west (Andes lie in Peru/Bolivia, outside Brazil), all major rivers of Brazil flow towards the north or east into the Atlantic Ocean, and hence there are no west-flowing rivers within Brazil.

(2) Dissimilarities between eastern and western coasts of India:

The western coast is narrow and rocky; spurs from the Western Ghats extend up to the coast, and rivers here are short and swift, forming estuaries, not deltas. The eastern coast, in contrast, is broad and has formed due to depositional work of long, gently-flowing rivers (Godavari, Krishna, Kaveri, Mahanadi) that form deltas as they meet the Bay of Bengal. Hence the two coasts differ in width, formation and river mouth types.

(3) Northeastern part of Brazilian Highlands receives very less rainfall:

The Great Escarpment acts as a barrier to the South-East Trade winds blowing from the Atlantic Ocean, causing them to shed their moisture on the windward (coastal) side. By the time these winds cross the escarpment and reach the northeastern part of the highlands, they become dry, creating a rain-shadow region known as the 'Drought Quadrilateral', which receives very little rainfall.

(4) Tropical cyclones occur rarely in Brazil:

Near the equator along the Brazilian coast, the temperature differences are negligible and winds move in a vertical direction rather than a strong horizontal/rotational one. The convergence zone of the trade winds here is weak, which does not allow the strong low-pressure rotational systems needed for cyclone formation to develop. Hence, tropical cyclones rarely visit the coasts of Brazil.

Q.4 Answer the following (Any Two): [3 marks each]

(1) Note on the Great Escarpment of Brazil:

Though it occupies a small area, the Great Escarpment forms a distinct physiographic region on the eastern edge of the Brazilian Highlands due to its steep slope and its effect on climate. Its average altitude is about 790 m, and it is particularly steep between Sao Paulo and Porto Alegre. It acts as a barrier to the South-East Trade winds, causing orographic rainfall on the coastal (windward) side, while creating a rain-shadow area called the 'Drought Quadrilateral' to its northeast, in the interior of the highlands.

(2) Differentiate between physiography of Brazil and India:

India has five major physiographic divisions: the Himalayas (young fold mountains), the North Indian Plains, the Peninsula (with old fold mountains like the Aravalis, plateaus, Vindhya-Satpuda ranges, Western and Eastern Ghats), Coastal Plains, and Island groups.

Brazil has five major physiographic divisions: the Highlands (Guyana Highlands and Brazilian Highlands/Shield), the Great Escarpment, the Coastal region, the Plains (Amazon Basin and Paraguay-Parana source region), and Island groups.

Unlike India, Brazil has no high young fold mountains like the Himalayas; most of Brazil is occupied by ancient plateaus/highlands, and its plains (Amazon Basin) are far larger in extent relative to mountainous terrain than India's North Indian Plains.

(3) Compare the climates of Brazil and India:

India has a monsoon type of climate, with distinct seasons (hot weather, rainy/monsoon, retreating monsoon, cold weather) and a large range of temperature between summer and winter, especially in the north. Rainfall in India comes mainly from the South-West and North-East Monsoon winds, and is highly seasonal.

Brazil, due to its vast latitudinal extent, experiences hot equatorial climate near the equator and temperate climate near the Tropic of Capricorn, with rainfall received throughout the year in equatorial regions from the South-East and North-East Trade winds. Unlike India, Brazil's climate shows negligible temperature variation near the equator, does not have a true 'monsoon', and rarely experiences tropical cyclones, whereas India regularly faces cyclones, especially on its eastern coast.